

Plan

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- Elementary Matrix
- Gauss-Jordan Method for Computing Inverse

The Inverse of a Matrix

An $n \times n$ matrix A is said to be **invertible** if there exists a matrix B satisfying $AB = I_n = BA$, and B is called an **inverse** of A .

- For example, the matrix $A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$ is invertible since

$$\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}.$$

- We can talk of **invertibility** only for square matrices.
- The zero matrix $\mathbf{O}$ is **not invertible**.
- The matrix $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ is **not** invertible, we will see why.

Result

If A is an invertible matrix, then its inverse is *unique*.

Proof. If B and C are inverses of A , then

$$B = BI_n = B(AC) = (BA)C = I_n C = C.$$

- A^{-1} denotes the inverse of an invertible matrix A .
- If A is invertible then $AA^{-1} = I_n = A^{-1}A$.
- When is a 1×1 matrix A invertible? What is A^{-1} then?

Result

If A is an invertible matrix, then

- 1 A^{-1} is also invertible, and $(A^{-1})^{-1} = A$;
- 2 for $c \neq 0$, cA is also invertible, and $(cA)^{-1} = \frac{1}{c}A^{-1}$;
- 3 A^t is invertible, and $(A^t)^{-1} = (A^{-1})^t$.
- 4 for $k \in \mathbb{N}$, A^k is invertible, and $(A^k)^{-1} = (A^{-1})^k$.

Result

The matrix AB is invertible, and $(AB)^{-1} = B^{-1}A^{-1}$.

Result

Suppose an $n \times n$ matrix A is invertible. Then

- $A\mathbf{x} = \mathbf{b}$ has a *unique solution* for each $\mathbf{b} \in \mathbb{F}^n$.
- In particular, $A\mathbf{x} = \mathbf{0}$ has only the *trivial solution*.

Result

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

- If $ad - bc \neq 0$, then A is invertible, and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

- If $ad - bc = 0$, then A is *not* invertible.

Because, in that case rows of A are *linearly dependent*, and so, $A\mathbf{x} = \mathbf{0}$ has *infinitely many solutions*.

Elementary Matrix:

An **elementary matrix** is a matrix that can be obtained by performing **an** elementary row operation on the identity matrix.

- There are **three types** of elementary matrices.
- For example, the following are examples of the three types of elementary matrices of size 3.

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}.$$

- The matrix E_1 is obtained by performing $R_2 \leftrightarrow R_3$ on I_3 .
- E_2 is obtained by performing $R_2 \rightarrow 5R_2$ on I_3 .
- E_3 is obtained by performing $R_3 \rightarrow R_3 + 2R_1$ on I_3 .

• Let $A = \begin{bmatrix} a & b & c \\ x & y & z \\ p & q & r \end{bmatrix}$. Then

$$E_1 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ x & y & z \\ p & q & r \end{bmatrix} = \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix}.$$

Similarly,

$$E_2 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} A = \begin{bmatrix} a & b & c \\ 5x & 5y & 5z \\ p & q & r \end{bmatrix} \text{ and}$$

$$E_3 A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} A = \begin{bmatrix} a & b & c \\ x & y & z \\ p+2a & q+2b & r+2c \end{bmatrix}.$$

- The matrix E_1A is the matrix obtained from A by performing the elementary row operation $R_2 \leftrightarrow R_3$.
- The matrix E_2A is the matrix obtained from A by performing the elementary row operation $R_2 \rightarrow 5R_2$.
- The matrix E_3A is the matrix obtained from A by performing the elementary row operation $R_3 \rightarrow R_3 + 2R_1$.

Q: Should these hold only when A is a square matrix? **NO.**

Result

Let E be an *elementary matrix* obtained by an elementary row operation on I_n . If the same *elementary row operation* is performed on an $n \times r$ matrix A , then the resulting matrix is equal to EA .

Result

The matrices A and B are row equivalent *if and only if* there are elementary matrices $E_1, E_2, \dots, E_k$ such that

$$B = E_k \cdots E_2 E_1 A.$$

- Elementary matrices are **always invertible**.

Applying $R_2 \leftrightarrow R_3$ on E_1 we get I_3 . Therefore,

$$E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = E_1.$$

Applying $R_2 \rightarrow \frac{1}{5}R_2$ on E_2 we get I_3 , and

applying $R_3 \rightarrow R_3 - 2R_1$ on E_3 we get I_3 .

Therefore, we have

$$E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}.$$

Result

Every elementary matrix is *invertible*, and its inverse is an elementary matrix of the *same type*.

Result

Let A and B be row equivalent, and $B = E_k \cdots E_2 E_1 A$.

- Then $A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} B$.
- A is invertible if and only if B is invertible.

Suppose A is $n \times n$ and is invertible. Let $B = RREF(A)$. Then B is invertible. (B is row equivalent to A .)

Can B have a zero row? If so, then $I_n = BB^{-1}$ has zero row! What is B then? B is $n \times n$, in *RREF* and has no zero rows.

So, $B = I_n$. Thus, A is a product of elementary matrices.

Result

Let A be an $n \times n$ matrix. Then

$$A \text{ is invertible} \iff RREF(A) = I_n \iff rank(A) = n.$$

Result (Fundamental Theorem of Invertible Matrices: I)

Let A be an $n \times n$ matrix. Then the following statements are equivalent.

- 1 A is *invertible*.
- 2 A^t is *invertible*.
- 3 $A\mathbf{x} = \mathbf{b}$ has a *unique solution* for every $\mathbf{b}$ in $\mathbb{R}^n$.
- 4 $A\mathbf{x} = \mathbf{b}$ has a *solution* for every $\mathbf{b}$ in $\mathbb{R}^n$.
- 5 $A\mathbf{x} = \mathbf{0}$ has only the *trivial solution*.
- 6 $\text{rank}(A) = n$.
- 7 The reduced row echelon form of A is I_n .
- 8 The rows of A are linearly independent.
- 9 The columns of A are linearly independent.
- 10 A is a *product of* elementary matrices.

Result

Let A be a square matrix. If B is a square matrix such that either $AB = I$ or $BA = I$, then A is invertible and $B = A^{-1}$. In particular, $AB = I \implies BA = I$.

Proof. Suppose $AB = I$. Then

$$B\mathbf{x} = \mathbf{0} \implies I\mathbf{x} = AB\mathbf{x} = A\mathbf{x} = \mathbf{0} \implies \mathbf{x} = \mathbf{0}.$$

So, B is invertible, and $A = A(BB^{-1}) = (AB)B^{-1} = IB^{-1} = B^{-1}$.
Note: Then $BA = BB^{-1} = I$.

Result

Let A be a square matrix. Then

$$E_k \cdots E_2 E_1 A = I \implies E_k \cdots E_2 E_1 I = A^{-1},$$

that is, if a finite sequence of elementary row operations transforms A to the identity matrix I , then the same sequence of elementary row operations transforms I to A^{-1} .

Gauss-Jordan Method for Computing Inverse

Let A be an $n \times n$ matrix.

- Apply elementary row operations on the matrix $[A \mid I_n]$.
- If A is invertible, then $[A \mid I_n]$ will be transformed to $[I_n \mid A^{-1}]$.
- If A is not invertible, then $[A \mid I_n]$ can **never** be transformed to a matrix of the type $[I_n \mid B]$.

Example

Find the inverse of the following matrix A , if it exists:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$